Predicting Algal Blooms with High Resolution Imaging & Machine Learning

Adams Yang
ady005@ucsd.edu

Sohan Raval
sraval@ucsd.edu

Owen Yang
okyang@ucsd.edu

Sivaram Ari
sari@ucsd.edu

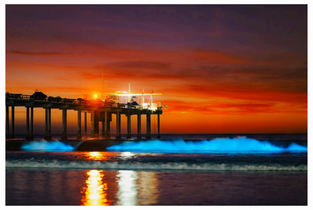
Mentor:
Duncan Watson-Parris
dwatsonparris@ucsd.edu

UC San Diego
HALICIOĞLU DATA SCIENCE INSTITUTE

1. Introduction

First Look

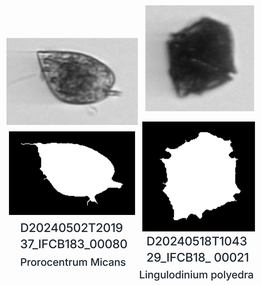
- Algal blooms are rapid proliferation of phytoplankton population
  - Linked to nutrient enrichment (eutrophication) and climate change.
- Harmful blooms degrade ecosystems and threaten coastal economies costing hundreds of millions in damages (Anderson et al., 2000).
- Target species *Lingulodinium polyedra* shown to cause red tides and produce harmful yesstoxins affecting marine life, like shellfish.
  - L. Polyedra produces bioluminescence seen at Scripps Pier (La Jolla, CA) during red tides.



@erikjepsen

Description of Data

- High-resolution imaging using Imaging FlowCytobot (IFCB) at Scripps Institute of Oceanography (SIO).
- Samples ~5 mL of seawater every ~20 minutes using automated optical microscopy.
- Derives morphological and texture-based features from each imaged taxa.
- Web scraping, 2022-2025, ~804 days, 120k files, 41,306 sampling bins.
- SCCOOS and NOAA Tides & Currents (Scripps Pier)
- Provide high-resolution physical and chemical ocean observations



McLane labs

2. Methods

Images → Feature Extraction Classification → Time Series Construction → Empirical Dynamic Modeling → Bloom Forecasting

Empirical Dynamic Modeling

- Time series normalized and imputed using EMA (30-day window).
- Delay embedding with dimension E = 4 and lag τ = 1–30 days

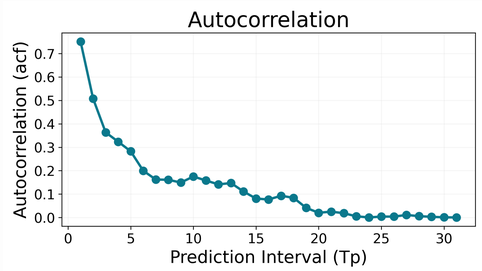
$$x(t)=[x(t),x(t-\tau),x(t-2\tau),x(t-3\tau)]$$

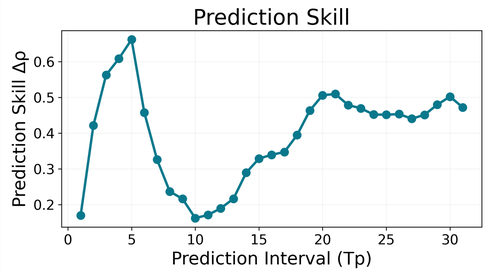
- Evaluated using Pearson correlation (ρ), RMSE, and MAE between observed and predicted values.
- Validation using leave-one-out cross-validation & IFCB 158.
- Baseline comparison: Simplex Projection and ARIMA,
- Multiview Embedding
  - Each embedding included abundance + 3 random features.
  - 500 feature combinations tested per prediction lag (Tp)

3. Results

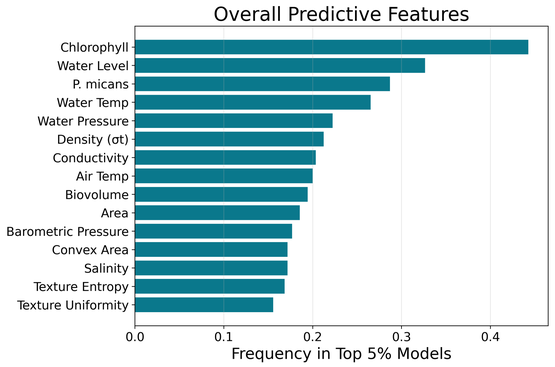
Predictive Skill Beyond Autocorrelation

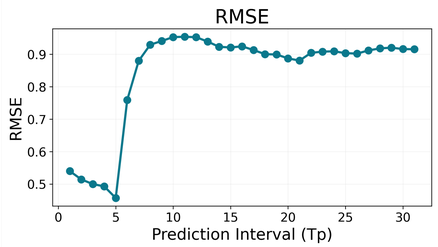
- Multiview embedding produced strong short-term forecast skill.





- Prediction shows strong short-term autocorrelation, with ACF ≈ 0.75 at Tp = 1 day.
- Autocorrelation decreases rapidly over time, dropping below 0.3 after ~5 days and approaching zero by ~20 days.
- Predictive skill peaked at 5-day prediction horizon, decreasing for mid-range horizons (6–12 days) but remains moderate at longer timescales (20–30 days).

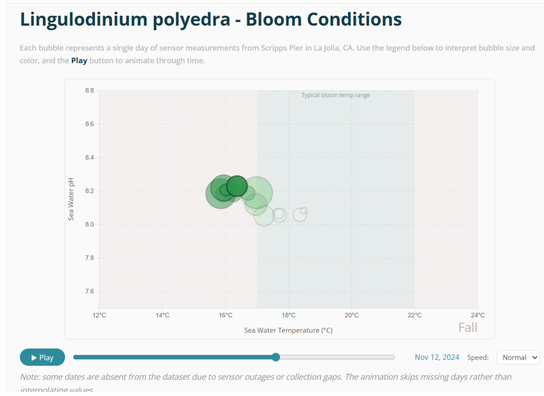




- Importance estimated from the top 5% of multiview embedding models, averaged across prediction horizons (Tp = 1–30 days).
- Chlorophyll concentration was the strongest predictor, consistent with its role as a key algal bloom indicator.
- Physical variables (water level, temperature, salinity) and image-derived plankton features provided additional predictive signal.
- Combining environmental and biological variables improved bloom forecasting.

Visualizing Algal Blooms

- An interactive bubble chart to explore 804 days of plankton and environmental observations from Scripps Pier at La Jolla (Oct 2022–Dec 2025).
- Each point represents a single day, encoding four variables:
  - Temperature (x-axis)
  - pH (y-axis)
  - L. polyedra concentration (bubble size)
  - Chlorophyll concentration (bubble color)



- Bloom events cluster within 17–22 °C temperature range, consistent with temperature as a predictor for blooms.
- Elevated pH values reflect increased photosynthetic activity.
- High chlorophyll concentrations frequently precede major blooms, consistent with prior works.
- Temporal animation shows bloom events develop rapidly over ~2–4 days, consistent with the short-term predictability detected by EDM models.

4. Conclusion

- Image-derived phytoplankton features and environmental data provide predictive signals for Lingulodinium polyedra bloom dynamics.
- Multiview embedding improves prediction by capturing nonlinear ecosystem interactions across multiple image and environmental features.

Limitations

- Long-term monitoring datasets may contain gaps or irregular sampling, requiring interpolation.
- Fixed sampling locations may miss regional bloom dynamics; multiple IFCB systems could improve spatial forecasting.

5. References

- Agarwal et al., 2023. Sub-monthly prediction of harmful algal blooms using automated cell imaging.
- Özlem Baydaroğlu. 2025. "Harmful algal bloom prediction using empirical dynamic modeling."
- Sosik et al. 2007. "A submersible imaging-in-flow instrument to analyze nano-and microplankton: Imaging FlowCytobot."
- Szewczyk et al. 2025. "Ensemble models improve near-term forecasts of harmful algal bloom and biotoxin risk."

Scan Me !